MATH9102

COMPARING NOMINAL VARIABLES

Sources used in creation of this lecture:
Statistics and Data Analysis, Peck, Olsen and Devore; Discovering Statistics Using IBM SPSS, Andy Field; Understanding Basic Statistics, Brase and Brase;SPSS Survival Manual, Julie Pallant

Comparing Nominal Variables

When two nominal variables are independent, there is no relationship between them.

This means that

- For all cases
  - The classification of a case into a particular category of one variable (the group variable) has no effect on the probability that the case will fall into any particular category of the second variable (the test variable).

`MATH9102-W4-Nominal.qmd`

Comparing Nominal Variables

Analyzing categorical variables

- Central Tendency (mean/median) of a categorical variable is meaningless
  - The numeric values you attach to different categories are arbitrary
  - The mean of those numeric values will depend on how many members each category has.
- Therefore, we analyze frequencies.

Comparing Nominal Variables

Using the **bullying.dat** dataset (Paul Connolly)

- Dataset Descriptor:
- https://www.routledge.com/Quantitative-Data-Analysis-in-Education-A-Critical-Introduction-Using-SPSS/Connolly/p/book/9780415372985
- Contains the results of national school level survey, subset of this survey which deals with bullying.

Question:

- "Is there a relationship between being bullied in school and participating in bullying of others in school?"
  - H0: There is no relationship between being a victim of bullying in school and participating in bullying others in school.
  - HA: There is a relationship between being a victim of bullying in school and participating in bullying others in school.
- Variables:
- 'ubullsch' – Q26. Have you yourself ever been bullied in school? {1- Yes, 2 –No, 99-not answered}
- 'ubulloth' - Q28 - Have you yourself ever taken part in bullying other students? {1- Yes, 2 –No, 99-not answered}

Comparing Nominal Variables

We could create a contingency table/
A → Students who were bullied and bullied others
B → Students who were bullied but did not bully others
C → Students who were not bullied but bullied others
D → Students who were not bullied and did not bully others
N → Total number of students

A Chi square test will compare the observed frequencies (A, B, C, D) with the expected frequencies under the assumption being bullied and bullying others are independent.

The chi-square statistic will be calculated, and its significance will be determined based on the p-value and the degrees of freedom (df = 1 in this case).

	Bullied Others	Did Not Bully Others	Total
Was Bullied	A	B	A + B
Was Not Bullied	C	D	C + D
Total	A + C	B + D	N

Chi-square Test

We are comparing observed values to expected values according to a specific hypothesis (null hypothesis)

The null hypothesis for this test states that the proportions (the distribution across categories) are the same for all of the populations

Views the data as two (or more) separate samples representing the different populations being compared

- The same variable is measured for each sample by classifying individual subjects into categories of the variable.
- The data are presented in a matrix with the different samples defining the rows and the categories of the variable defining the columns.
- The data, called **observed frequencies**, simply show how many individuals from the sample are in each cell of the matrix.

	Bullied Others	Did Not Bully Others	Total
Was Bullied	A	B	A + B
Was Not Bullied	C	D	C + D
Total	A + C	B + D	N

Chi Square Test for Independence

We are comparing observed values to **expected values** according to a specific hypothesis (null hypothesis)

- The null hypothesis for this test states that the proportions (the distribution across categories) are the same for all of the populations
- The null hypothesis is used to construct an idealized sample distribution of **expected frequencies** that describes how the sample would look if the data were in perfect agreement with the null hypothesis
- Calculated from the observed frequencies:

$$E = \frac{(Row\,Total) \times (Column\,Total)}{Grand\,Total}$$

Expected Frequencies

Suppose we had the following observed frequencies:

	Bullied Others	Did Not Bully Others	Total
Was Bullied	A = 35	B = 209	A + B = 244
Not Bullied	C = 25	D = 530	C + D = 555
Total	A + C = 60	B + D = 739	**N = 799**

	Formula	Expected Frequency
E(A) = "Was Bullied" & "Bullied Others"	(244 × 60) / 799	**18.32**
E(B) = "Was Bullied" & "Did Not Bully"	(244 × 739) / 799	**225.68**
E(C) = "Not Bullied" & "Bullied Others"	(555 × 60) / 799	**41.68**
E(D) = "Not Bullied" & "Did Not Bully"	(555 × 739) / 799	**513.32**

Comparing

	Bullied Others	Did Not Bully Others	Total
Was Bullied	A = 35	B = 209	A + B = 244
Not Bullied	C = 25	D = 530	C + D = 555
Total	A + C = 60	B + D = 739	**N = 799**

	Formula	Expected Frequency
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E(B) = "Was Bullied" & "Did Not Bully"	(244 × 739) / 799	**225.68**
E(C) = "Not Bullied" & "Bullied Others"	(555 × 60) / 799	**41.68**
E(D) = "Not Bullied" & "Did Not Bully"	(555 × 739) / 799	**513.32**

Comparing these to the observed counts (35 vs. 18.32, 25 vs. 41.68, etc.) shows a clear deviation — especially:

- More students than expected were both **bullied and bullied others (A)**.
- Fewer students than expected were **not bullied but bullied others (C)**.

This is what our chi square test will measure and will test to see if the difference is large enough to be statistically significant.

Chi-Square Test for Independence

A chi-square statistic is computed to measure the amount of discrepancy between the ideal sample (expected frequencies from H_0) and the actual sample data (the observed frequencies).

The output is a cross-tabulation table

In R

```
#CrossTable(predictor, outcome, fisher = TRUE, chisq = TRUE, expected =
TRUE)#Note: for this approach there must be the same number of values for
both variables so#missing data is important.

# Remove rows where either variable is NA

bully_clean <- bully %>%  filter(!is.na(ubullsch) &
!is.na(ubulloth))

# Now run CrossTable requesting both Fisher and Chi Square so we have full
information# We request the format in SPSS to make it easier to interpret.


gmodels::CrossTable(
bully_clean$ubullsch,
bully_clean$ubulloth,
fisher = TRUE,
chisq = TRUE,
expected = TRUE,
sresid = TRUE,
format = "SPSS")
```

```
   Cell Contents
|-------------------------|
|                   Count |
|         Expected Values |
| Chi-square contribution |
|             Row Percent |
|          Column Percent |
|           Total Percent |
|            Std Residual |
|-------------------------|

Total Observations in Table:  799

                     | bully_clean$ubulloth
bully_clean$ubullsch |        No  |       Yes  | Row Total |
---------------------|-----------|-----------|-----------|
                  No |       530 |        25 |       555 |
                     |   513.323 |    41.677 |           |
                     |     0.542 |     6.673 |           |
                     |   95.495% |    4.505% |   69.462% |
                     |   71.719% |   41.667% |           |
                     |   66.333% |    3.129% |           |
                     |     0.736 |    -2.583 |           |
---------------------|-----------|-----------|-----------|
                 Yes |       209 |        35 |       244 |
                     |   225.677 |    18.323 |           |
                     |     1.232 |    15.179 |           |
                     |   85.656% |   14.344% |   30.538% |
                     |   28.281% |   58.333% |           |
                     |   26.158% |    4.380% |           |
                     |    -1.110 |     3.896 |           |
---------------------|-----------|-----------|-----------|
        Column Total |       739 |        60 |       799 |
                     |   92.491% |    7.509% |           |
```

Make sure your %s add up to 100.

```
Pearson's Chi-squared test
------------------------------------------------------------
Chi^2 =  23.62668     d.f. =  1     p =  1.169546e-06

Pearson's Chi-squared test with Yates' continuity correction
------------------------------------------------------------
Chi^2 =  22.2312     d.f. =  1     p =  2.417134e-06


Fisher's Exact Test for Count Data
------------------------------------------------------------
Sample estimate odds ratio:  3.543753

Alternative hypothesis: true odds ratio is not equal to 1
p =  3.65894e-06
95% confidence interval:  2.006207 6.341761

Alternative hypothesis: true odds ratio is less than 1
p =  0.9999992
95% confidence interval:  0 5.799364

Alternative hypothesis: true odds ratio is greater than 1
p =  3.154022e-06
95% confidence interval:  2.184139 Inf


   Minimum expected frequency: 18.3229
```

Need to check this – assumption is that lowest expected frequency is 5 or more. If not then this test is not appropriate and should use Fisher's Exact Probability Test.

If we are looking at 2x 2 table we look at Yates continuity correction

```
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95% confidence interval:  2.184139 Inf
```

Yate’s Continuity Correction

This is to help prevent us making a Type I error

When we are doing a Chi-square test, we are assuming that the probability distribution of the binomial frequencies observed approximate the Chi-square distribution

- This is not quite correct
- Yate’s correction adjusts Pearson’s Chi-square by subtracting 0.5 from each of the differences between the observed values and the expected values, which will reduce the Chi-square statistic and its associated p-value
- Particularly important for small data (larger datasets this can be overcome)
- May tend to overcorrect – make sure you reflect your research area’s perspective (some would say Yates is unnecessary even for small data).

In R – more simplistic

```
#Create your contingency table

mytable<-xtabs(~ubullsch+rsex, data=bully)


ctest<-chisq.test(mytable, correct=TRUE)#chi square test

#correct=TRUE to get Yates correction needed for 2x2 table


ctest#will give you the details of the test statistic and p-value

ctest$expected#expected frequencies

ctest$observed#observed frequencies

ctest$p.value

OUTPUT:

X-squared = 2.3005, df = 1, p-value = 0.1293
```

Chi Square Statistic

Measures the amount of discrepancy between the ideal sample (expected frequencies from H_0) and the actual sample data (the observed frequencies = f_o).

- A large discrepancy results in a large value for chi-square which indicates that the data do not fit the null hypothesis and the hypothesis does not hold.
- We compare our test statistic to the Chi Square distribution (in the form of a table) and we can get the probability that any deviation in the observed from the expected is due to chance only

For one degree of freedom

- The critical value associated with $p = 0.05$ for Chi Square is 3.84
- The critical value associated with $p = 0.01$ it is 6.64
- *Chi-square values higher than this critical value are associated with a statistically low probability that H_0 holds.*

Effect Sizes

Phi (2 x 2 contingency table)

$$\varphi = \sqrt{\frac{\chi^2}{n}}$$

0.1 is considered a small effect

0.3 a medium effect

0.5 a large effect.

Phi is used for 2×2 tables

Effect Sizes

Cramers V depends on the degrees of freedom

Cramer’s V is suitable for tables larger than 2×2, and

$$V = \sqrt{\frac{\chi^2}{n \cdot df^*}}$$

df	*small*	*medium*	*large*
1	.10	.30	.50
2	.07	.21	.35
3	.06	.17	.29
4	.05	.15	.25
5	.04	.13	.22

Chi-Square Test for Independence

Reporting the findings

A Chi-square test of independence was conducted to examine the relationship between being bullied in school and bullying others. The association was statistically significant ($\chi^2(1, N = 799) = 22.23, p < .001$). Students who reported being bullied were significantly more likely to report bullying others (14.3%) than those who were not bullied (4.5%). The effect size, measured by Phi ($\phi = 0.17$), indicates a small association. There is therefore evidence to support rejecting the null hypothesis in favour of the alternate hypothesis.

Where did the %s come from?

```
                     | bully_clean$ubulloth 
bully_clean$ubullsch |        No |       Yes | Row Total | 
---------------------|-----------|-----------|-----------|
                  No |       530 |        25 |       555 | 
                     |   513.323 |    41.677 |           | 
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---------------------|-----------|-----------|-----------|
        Column Total |       739 |        60 |       799 | 
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```

Bullied students who bullied others:

- (35/244)×100=14.3%

Bullied students who did not bully others:

- (209/244)×100=85.7%

Where did the %s come from?

bully_clean$ubullsch	bully_clean$ubulloth No	Yes	Row Total
No	530	25	555
	513.323	41.677	
	0.542	6.673	
	95.495%	4.505%	69.462%
	71.719%	41.667%	
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Yes	209	35	244
	225.677	18.323	
	1.232	15.179	
	85.656%	14.344%	30.538%
	28.281%	58.333%	
	26.158%	4.380%	
	-1.110	3.896	
Column Total	739	60	799
	92.491%	7.509%	

Non-Bullied students who bullied others:

- (25/555)×100=4.5%

Non-Bullied students who did not bully others:

- (530/555)×100=95.5%

Repeated Measures Categorical Variables

Use McNemar’s test

Matched Samples or repeated measures (pre-test/post-test)

- Two variables one recorded at Time 1 and one recorded at Time 2 (after an intervention)

Use for categorical variables with two response options

- For 3 or more use Cochran’s Q-test

In R: mcnemar.test

Repeated Measures Categorical Variables

You can try this by using one of Pallant's Dataset

- experiment.dat
- Fields Time1 Clinical Depression, Time 2 Clinical Depression
- You should get a conclusion that indicates there is no significant change in those diagnosed as clinically depressed after the program of treatment